



GOSFORD HIGH SCHOOL

2007

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

MATHEMATICS EXTENSION 1

General Instructions:

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- Each question should be started on a new page.
- All necessary working should be shown in every question

Total marks: - 84

- Attempt Questions 1 -7
- All questions are of equal value

Question 1: (12 marks)

(a) Solve $\frac{2x}{x-3} \leq 1, x \neq 3$. (3)

(b) Find $\frac{d}{dx}(x \cos^{-1} x)$. (2)

(c) If α, β, γ are the roots of $2x^3 + 4x^2 - 6x + 3 = 0$ find:

(i) $\alpha + \beta + \gamma$. (1)

(ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. (2)

(d) The interval joining the points A (2,1) and B (-3, -4) cuts the x-axis at C. Find the ratio in which the point C divides AB. (4)

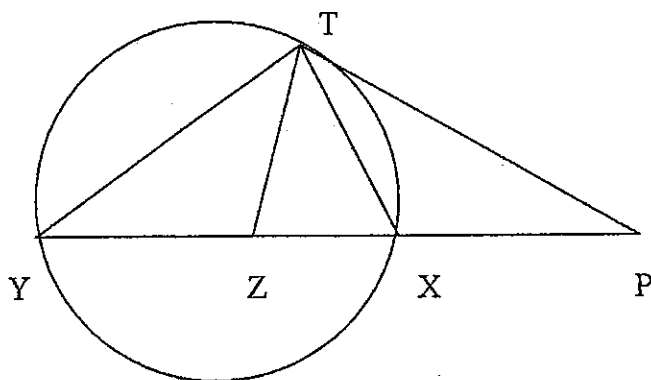
Question 2: (12 marks)

(a) Evaluate $\int_0^{\sqrt{2}} \frac{dx}{\sqrt{2-x^2}}$ (2)

(b) Use the substitution $u = 1+x$ to find $\int_0^1 \frac{x}{\sqrt{1+x}} dx$ in simplest surd form. (4)

(c) The acute angle between the lines $2x - y + 1 = 0$ and $kx + y + 5 = 0$ is 45° . Find the value(s) of k . (3)

(d) PT is a tangent to the circle and PXY is a secant. Z is a point on PXY such that $TX = TZ$. Prove that $\angle YTZ = \angle XPT$. (3)



NOT
TO
SCALE

Question 3: (12 marks)

(a) Prove that $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$ (2)

(b) (i) Draw a neat sketch of $y = 2 \sin^{-1} 3x$ and state its domain and range. (2)

(ii) Find the volume of the solid of revolution obtained by rotating the curve $y = 2 \sin^{-1} 3x$ about the y-axis between $y=0$ and $y = \frac{\pi}{2}$. (4)

(c) Prove by the principle of Mathematical Induction that

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}. \quad (4)$$

Question 4: (12 marks)

(a) Show that $\frac{dv}{dt} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ (1)

(b) A particle moves in a straight line so that its acceleration 'a' in m/s^2 is given by $a = -e^{\frac{-x}{2}}$. When $t = 0$, $x = 0$ and $v = 2m/s$.

(i) Show that the velocity of the particle is given by $v = 2e^{\frac{-x}{4}}$. (3)

(ii) Find the displacement and velocity of the particle in terms of 't'. (3)

(c) The point $P(2p, p^2)$ lies on the parabola $x^2 = 4y$ whose focus is S .

(i) Find the equation of the tangent to the parabola at P . (2)

(ii) The tangent at P meets the x-axis at Q . The point R divides the interval joining SQ externally in the ratio 3: 2. Show that as P moves on the parabola $x^2 = 4y$ the locus of R is a straight line. (3)

Question 5: (12 marks)

(a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$. (1)

(b) Find the general solution to $\cos 2\theta = \cos \theta$. (3)

(c) For the function $f(x) = e^{x-1}$ find the inverse function $f^{-1}(x)$ and hence

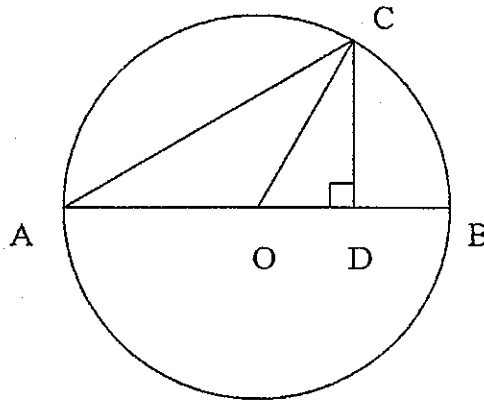
show that $f[f^{-1}(x)] = f^{-1}[f(x)] = x$. (3)

(d) (i) The equation $x^3 - kx^2 + 2 = 0$ has exactly one root between $x=0$ and $x=1$. Prove that $k > 3$. (2)

- (ii) The equation $x^3 - 4x^2 + 2 = 0$ has a root between $x=0$ and $x=1$. Using a first approximation of this root as $x=0.7$, use one application of Newton's Method to find a better approximation correct to 2 decimal places. (3)

Question 6: (12 marks)

- (a) AB is a diameter of a circle centre O, radius 2 units. $CD \perp AB$.

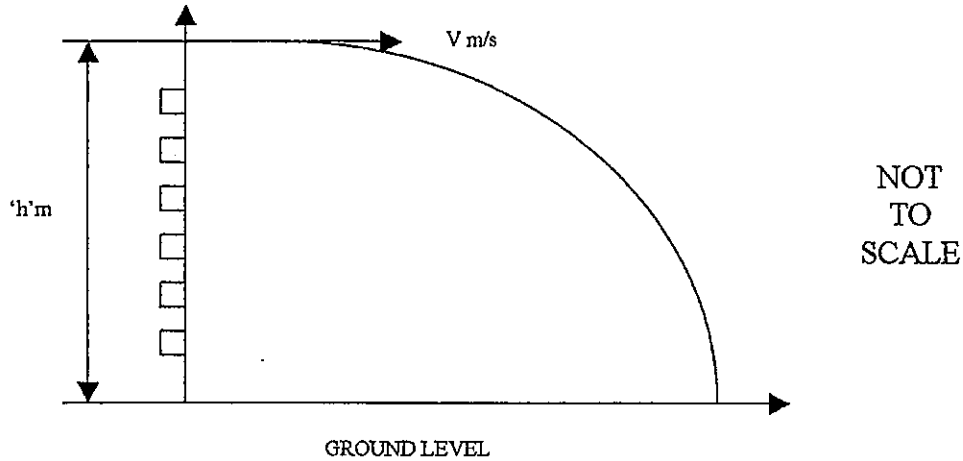


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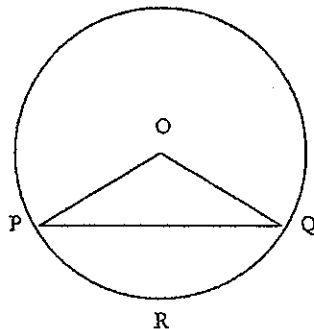
- (i) If $\angle CAO = \theta$, explain why $\angle DOC = 2\theta$. (1)
- (ii) Show that the perimeter 'P' of $\triangle DOC$ is given by:
 $P = 2 + 2(\sin 2\theta + \cos 2\theta)$. (2)
- (iii) Express $\sin 2\theta + \cos 2\theta$ in the form $R \sin(2\theta + \alpha)$ where α is acute. (2)
- (iv) Find the maximum value of the perimeter 'P' and the value of θ for which the perimeter is a maximum. (2)
- (b) A particle moving in Simple Harmonic Motion starts at the centre of oscillation O with an initial velocity of 3 cm/s . If it has a period of $\frac{\pi}{2}$ seconds, find:
- (i) the value of 'n'. (1)
- (ii) the amplitude 'a' of the motion. (1)
- (iii) the time taken for the particle to first reach a displacement of $x = -0.375 \text{ cm}$. (3)

Question 7: (12 marks)

- (a) An object is projected horizontally from the top of a building which is 'h' metres above ground level with a velocity of $V \text{ m s}^{-1}$. (Assume there is no air resistance and take the foot of the building to be the origin).



- (i) Show that the object strikes the ground when $t = \sqrt{\frac{2h}{g}}$ at a horizontal displacement of $V \sqrt{\frac{2h}{g}}$ metres from the foot of the building (2)
- (ii) Another object is projected horizontally from a height ' k ' times the height of the first object for some constant ' k '. If it is to strike the ground at the same point as the first object calculate the speed of projection in terms of ' V ' and ' k '. (2)
- (iii) If in part (i) the building is 40 metres high and the speed of projection is 20 m s^{-1} find the object's velocity when it strikes the ground and the angle at which it strikes the ground to the nearest degree. (Take $g = 10 \text{ m s}^{-2}$) (2)
- (b) The chord PQ cuts off a minor segment PRQ in a circle centre O radius ' r ' units. The angle POQ is ' x ' radians.



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It is given that 'r' and 'x' vary so that the area of the minor segment is constant at 50 cm^2 .

(i) Show that $r = \frac{10}{\sqrt{x - \sin x}}$. (2)

(ii) If 'x' is increasing at a rate of 0.05 rads/sec, find the rate at which the radius is decreasing when $x = 1.5$ rads correct to 2 decimal places. (4)

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1.

a) $\frac{2x}{x-3} \leq 1$

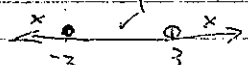
$x \neq 3$

If $\frac{2x}{x-3} = 1$

$2x = x-3$

$x = -3$

Since this is the solⁿ to the eqⁿ it can't be a solⁿ to the ineqⁿ.



By testing pts in the different regions on the no. line the solⁿ is

$-3 \leq x < 3$

b) Let $y = x \cos^{-1} x$

$\frac{dy}{dx} = v u' + u v'$

$= \cos^{-1} x \cdot 1 + x \cdot \frac{-1}{\sqrt{1-x^2}}$

$= \cos^{-1} x - \frac{x}{\sqrt{1-x^2}}$

c) $2x^3 + 4x^2 - 6x + 3 = 0$

(i) $\alpha + \beta + \gamma = -\frac{b}{a}$

$= -\frac{4}{2}$

$= -2$

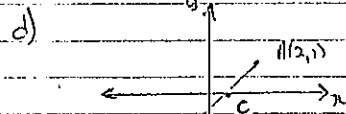
(ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

$= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$

$= \frac{c/a}{-d/a}$

$= \frac{-6/2}{-3/2}$

$= +2$



Gradient of AB is 1.

$\therefore$ Eqⁿ of AB is

$y - 1 = 1(x - 2)$

$y = x - 1$

If $y = 0$, $x = 1$

$\therefore C$ is the pt (1, 0)

Let the ratio be $k:1$

Using $x = \frac{kx_1 + x_2}{k+1}$

$1 = \frac{k \cdot 2 + k \cdot 3}{k+1}$

$1 = \frac{2-3k}{k+1}$

$k+1 = 2-3k$

$4k = 1$

$k = \frac{1}{4}$

$\therefore k:1 = \frac{1}{4}:1$

$\therefore$ the ratio is $1:4$

Question 2.

a) $\int_0^{\sqrt{2}} \frac{dx}{\sqrt{2-x^2}}$

$= \left[\sin^{-1} \left(\frac{x}{\sqrt{2}} \right) \right]_0^{\sqrt{2}}$

$= \sin^{-1} \left(\frac{\sqrt{2}}{\sqrt{2}} \right) - \sin^{-1} 0$

$= \frac{\pi}{2} - 0$

$= \frac{\pi}{2}$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$\therefore 1 = \left| \frac{2+k}{1-2k} \right|$

$\frac{2+k}{1-2k} = 1$ or $\frac{2+k}{1-2k} = -1$

$2+k = 1-2k$ or $2+k = -1+2k$

$3k = -1$ or $3 = k$

$k = -\frac{1}{3}$ or $k = 3$

b) If $u = 1+x$ $\frac{du}{dx} = 1$ $\frac{du}{u} = \frac{dx}{1+x}$

$\therefore \int_0^1 \frac{x}{1+x} dx = \int_1^2 \frac{u-1}{u} du$

$= \int_1^2 \left(1 - \frac{1}{u} \right) du$

$= \left[u - \ln u \right]_1^2$

$= \left(2 - \ln 2 \right) - \left(1 - \ln 1 \right)$

$= 2 - \ln 2 - 1 + 0$

$= 1 - \ln 2$

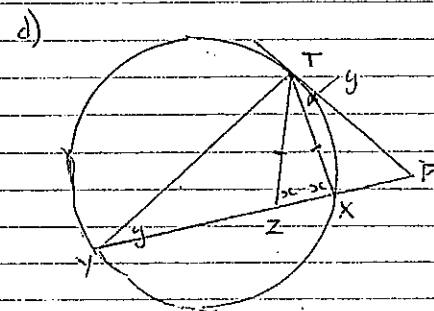
$= 1 - \ln 2$

$= \frac{4-2\sqrt{2}}{3}$

c) If $2x-y+1=0$, $y=2x+1$

If $kx+y+5=0$, $y=-kx-5$

$\therefore M_1 = 2 \neq M_2 = -k$



Let $\angle TXZ$ & $\angle TXZ$ be x°

(have L's of isosc ΔTXZ)

Let $\angle TYZ$ be y°

$\therefore \angle XTP = y^\circ$ (L in the alternate segment theorem)

$\angle YTZ = x^\circ - y^\circ$ (ext. L of a Δ theorem)

$\angle XPT = x^\circ - y^\circ$ (ext. L of a Δ theorem)

$\therefore \angle YTZ = \angle XPT$

Question 3.

a) $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$

LHS = $\frac{2 \sin \theta \cos \theta}{1 + 2 \cos^2 \theta - 1}$
 $= \frac{2 \sin \theta \cos \theta}{2 \cos^2 \theta}$ (2)

$= \frac{\sin \theta}{\cos \theta}$
 $= \text{RHS}$

$\therefore x^2 = \frac{1}{9} \sin^2 y$

$V = \pi \int_0^{\pi/2} \frac{1}{9} \sin^2 y \, dy$ (4)

$= \pi \int_0^{\pi/2} \frac{1}{9} \left(\frac{1}{2} - \frac{1}{2} \cos 2y \right) dy$

$= \pi \left[\frac{y}{2} - \frac{1}{4} \sin 2y \right]_0^{\pi/2}$

$= \pi \left\{ \left(\frac{\pi}{4} - \frac{1}{4} \sin \pi \right) - \left(0 - \frac{1}{4} \sin 0 \right) \right\}$

$= \pi \left(\frac{\pi}{4} - \frac{1}{4} \right)$

$= \frac{\pi^2}{36} - \frac{\pi}{18} \text{ units}^3$

c) Prove true for $n=1$

LHS = $\frac{1}{1 \times 3} = \frac{1}{3}$

RHS = $\frac{1}{2 \times 1 + 1} = \frac{1}{3}$

$\therefore$ True for $n=1$.

Assume $S(k)$ true

i.e. $\frac{1}{1 \times 3} + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2k+1)(2k+3)} = \frac{k}{2k+1}$

Prove $S(k+1)$ true

i.e. $\frac{1}{1 \times 3} + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2k+1)(2k+3)} + \frac{1}{(2k+1)(2k+3)} = \frac{k+1}{2k+3}$

LHS = $\frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)}$

$= \frac{2k^2 + 3k + 1}{(2k+1)(2k+3)}$

$= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)}$

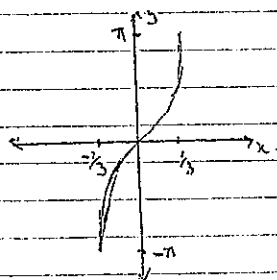
$= \text{RHS} \therefore \text{P.T.O.}$

(iii) $V = \pi \int_a^b x^2 \, dy$

If $\frac{y}{2} = \sin^{-1} 3x$

$3x = \sin \frac{y}{2}$

$x = \frac{1}{3} \sin \frac{y}{2}$



Therefore if the statement is true for $n=k$, it is true for $n=k+1$. Since it is true for $n=1$, it is true for $n=2$. Since true for $n=2$, it is true for $n=3$ and so on. $\therefore$ The statement is true for all positive integers n .

(ii) $\frac{dx}{dt} = \frac{2e^{-x/4}}{e^{x/4}}$

$\therefore \frac{dx}{dt} = \frac{2e^{-x/4}}{2}$

$t = \int \frac{1}{2} e^{x/4} dx$
 $= \int 2 \times \frac{1}{4} e^{x/4} dx$

$= 2e^{x/4} + k$

Question 4.

a) $\frac{dr}{dt} = \frac{dr}{dx} \times \frac{dx}{dt}$
 $= \frac{dr}{dx} \times \sqrt{1-v^2}$ (1)
 $= \frac{dr}{dx} \times \frac{d}{dr} \left(\frac{1}{2} v^2 \right)$
 $= \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

b) (i) $\frac{d}{dt} \left(\frac{1}{2} v^2 \right) = -e^{-x/2}$
 $\frac{1}{2} v^2 = \int -e^{-x/2} dx$
 $= 2e^{-x/2} + c$

When $x=0$, $v=2$.

$\therefore \frac{1}{2} (2^2) = 2e^0 + c$
 $2 = 2 + c$
 $c = 0$

$\therefore \frac{1}{2} v^2 = 2e^{-x/2}$ (3)
 $v^2 = 4e^{-x/2}$

$v^2 = 4e^{-x/2}$

$v = \pm 2e^{-x/4}$

$v = 2e^{-x/4}$ since $v=2$ when $x=0$.

ii) $t=0$, $x=0$
 $\therefore 0 = 2e^0 + k$
 $k = -2$

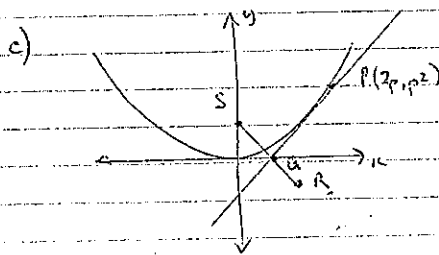
$t = 2e^{x/4} - 2$
 $t+2 = 2e^{x/4}$
 $\frac{t+2}{2} = e^{x/4}$ (3)

$\ln \left(\frac{t+2}{2} \right) = \frac{x}{4}$
 $x = 4 \ln \left(\frac{t+2}{2} \right)$

$x = 4 \left[\ln(t+2) - \ln 2 \right]$

$\frac{dx}{dt} = 4 \left[\frac{1}{t+2} - 0 \right]$

$v = \frac{4}{t+2}$



(i) If $x^2 = 4y$

$$y = \frac{x^2}{4}$$

$$\frac{dy}{dx} = \frac{2x}{4}$$

When $x = 2p$

$$m = \frac{dy}{dx}$$

$$= p$$

∴ Eqn of tangent is (2)

$$y - p^2 = p(x - 2p)$$

$$y - p^2 = px - 2p^2$$

$$y = px - p^2$$

(ii) When $y = 0$,

$$px - p^2 = 0$$

$$p(x - p) = 0$$

$$x = p$$

So O is the pt. $(p, 0)$

S: If $x^2 = 4ay$

$$x^2 = 4ay$$

$$a = 1$$

∴ So the pt. $(0, 1)$

$$S(0, 1) \text{ and } Q(p, 0)$$

$$3 = -2$$

$$x = -2 \times 0 + 3 \times p$$

$$= 3p$$

∴ P is the pt. $(3p, -2)$

∴ the locus of R is the straight line

$$y = -2$$

Question 5.

a) $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$

$$= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \frac{3}{5}$$

$$= 1 \times \frac{3}{5}$$

$$= \frac{3}{5}$$

(1)

b) $\cos 2\theta = \cos \theta$

$$2\cos^2\theta - 1 = \cos \theta$$

$$2\cos^2\theta - \cos \theta - 1 = 0$$

$$(2\cos \theta + 1)(\cos \theta - 1) = 0$$

$$\cos \theta = -\frac{1}{2} \text{ or } \cos \theta = 1$$

$$\theta = 2n\pi + \cos^{-1}(-\frac{1}{2}) \text{ or } 2n\pi + \cos^{-1}(1)$$

$$\theta = 2n\pi + \frac{2\pi}{3} \text{ or } 2n\pi$$

c) $f(x) = e^{x-1}$

$$\text{Let } y = e^{x-1}$$

$$\therefore x = e^{y-1}$$

$$\ln x = y - 1$$

$$y = \ln x + 1$$

$$\therefore f^{-1}(x) = \ln x + 1$$

$$f[f^{-1}(x)] = e^{\ln x + 1} = 1$$

$$= e^{\ln x}$$

$$= x$$

$$f^{-1}[f(x)] = \ln(e^{x-1}) + 1$$

$$= x - 1 + 1$$

$$= x$$

$$\therefore f[f^{-1}(x)] = f^{-1}[f(x)] = x$$

d) (i) $x^3 - kx^2 + 2 = 0$

If there is a root between 0 and 1

$$f(0) = f(1) \text{ must be opposite}$$

in sign

$$f(0) = 2 \text{ which is positive}$$

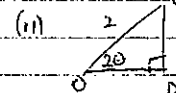
$$f(1) = 1 - k + 2$$

$$= 3 - k$$

$$\therefore 3 - k < 0$$

$$k > 3$$

(2)



$$OC = 2 \text{ units}$$

$$\frac{OA}{OC} = \cos 2\theta$$

$$\frac{AC}{OC} = \sin 2\theta$$

$$OA = 2 \cos 2\theta$$

$$AC = 2 \sin 2\theta$$

$$\therefore P = 2 + 2 \sin 2\theta + 2 \cos 2\theta$$

$$= 2 + 2(\sin 2\theta + \cos 2\theta)$$

(iii) $\sin 2\theta + \cos 2\theta = a \sin 2\theta + b \cos 2\theta$

$$\text{where } a = b = 1$$

(iii) $x^3 - 4x^2 + 2 = 0$

$$\text{Let } f(x) = x^3 - 4x^2 + 2$$

$$f'(x) = 3x^2 - 8x$$

$$a_2 = a_1 - \frac{f(a_1)}{f'(a_1)}$$

$$= 0.7 - \frac{f(0.7)}{f'(0.7)}$$

$$= 0.7 - \frac{0.383}{-4.13}$$

$$= 0.792736077$$

$$\approx 0.79 \text{ (2 d.p.)}$$

$$R = \sqrt{a^2 + b^2} = \sqrt{2}$$

$$\therefore \alpha = \frac{\pi}{4}$$

$$\therefore \sin 2\theta + \cos 2\theta = \sqrt{2} \left(\sin 2\theta + \frac{\pi}{4} \right)$$

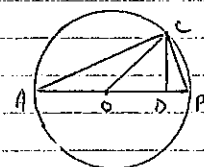
$$(iv) P = 2 + 2(\sin 2\theta + \cos 2\theta) = 2 + 2\sqrt{2} \left(\sin 2\theta + \frac{\pi}{4} \right)$$

$$\text{Max value of } \sin \left(2\theta + \frac{\pi}{4} \right) = 1$$

$$\therefore \text{Max } P = 2 + 2\sqrt{2} \times 1 = 2 + 2\sqrt{2} \text{ units}$$

Question 6.

a)



(i) If $\angle CAO = \theta$

$$\angle BOA = 2\theta$$

(angle at the center of a circle is twice the angle at the circumference)

If $\sin \left(2\theta + \frac{\pi}{4} \right) = 1$

$$2\theta + \frac{\pi}{4} = \frac{\pi}{2}$$

$$2\theta = \frac{\pi}{4}$$

$$\theta = \frac{\pi}{8}$$

(2)

(1)

i) Since the motion is SHM

$$T = \frac{2\pi}{n}$$

$$\therefore \frac{2\pi}{n} = \frac{\pi}{2}$$

$$n = 4$$

(iii) Since the motion is SHM

$$v^2 = n^2(a^2 - x^2)$$

$$\text{If } x=0, v=3, n=4$$

$$\therefore 9 = 16(a^2 - 0)$$

$$a^2 = \frac{9}{16}$$

$$a = \pm \frac{3}{4}$$

(1)

$$\therefore \text{Amplitude is } \frac{3}{4} \text{ cm.}$$

(ii) Let $x = a \sin(nt + \alpha)$

describe the motion.

$$\text{If } t=0, x=0,$$

$$\therefore 0 = a \sin(0 + \alpha)$$

$$\alpha = 0.$$

$$\text{So } x = \frac{3}{4} \sin 4t.$$

$$\text{When } x = -0.375$$

$$-0.375 = \frac{3}{4} \sin 4t$$

$$-1.5 = 3 \sin 4t$$

$$\sin 4t = -\frac{1}{2}$$

$$\therefore 4t = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$t = \frac{7\pi}{24}, \frac{11\pi}{24}$$

$\therefore$ The particle first reaches -0.375 cm after $\frac{7\pi}{24}$ seconds.

Question 7.

$$\text{a) (i) } \ddot{x} = 0$$

$$\dot{x} = \frac{16}{3}$$

$$x = \frac{16}{3}t$$

$$\ddot{y} = -g$$

$$\dot{y} = -gt$$

$$y = -\frac{1}{2}gt^2 + h$$

$$\text{If } y=0, 0 = -\frac{1}{2}gt^2 + h$$

$$\frac{1}{2}gt^2 = h$$

$$t^2 = \frac{2h}{g}$$

$$t = \pm \sqrt{\frac{2h}{g}}$$

(2)

$$\therefore t = \sqrt{\frac{2h}{g}} \text{ as } -\sqrt{\frac{2h}{g}} \text{ is meaningless}$$

$$\text{When } t = \sqrt{\frac{2h}{g}}$$

$$x = Vt$$

$$= V\sqrt{\frac{2h}{g}}$$

(iii) If object is fired from a height of kh

$$y = -\frac{1}{2}gt^2 + kh$$

$$\text{When } y=0, 0 = -\frac{1}{2}gt^2 + kh$$

$$\frac{1}{2}gt^2 = kh$$

$$gt^2 = 2kh$$

$$t^2 = \frac{2kh}{g}$$

$$t = \pm \sqrt{\frac{2kh}{g}}$$

$$\therefore t = \sqrt{\frac{2kh}{g}} \text{ as } -\sqrt{\frac{2kh}{g}} \text{ is meaningless.}$$

$$\text{ie } t = \sqrt{\frac{2h}{g}}$$

Let the speed be V .

$$\therefore X = V \sqrt{\frac{2h}{g}}$$

When $X = x$

$$V \sqrt{\frac{2h}{g}} = V \sqrt{\frac{2h}{g}}$$

$$V = \frac{V}{\sqrt{k}}$$

(2)

(iii) If $V=20, h=40, g=10$

$$x = 20t, y = -5t^2 + 40$$

$$\text{When } y=0, -5t^2 + 40 = 0$$

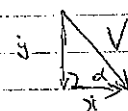
$$5t^2 = 40$$

$$t^2 = 8$$

$$t = \pm 2\sqrt{2}$$

$$\therefore t = 2\sqrt{2} \text{ as } -2\sqrt{2} \text{ is meaningless}$$

$$\therefore \dot{x} = 20, \dot{y} = -10t = -10 \cdot 2\sqrt{2} = -20\sqrt{2}$$



$$V^2 = (-20\sqrt{2})^2 + 20^2 = 800 + 400 = 1200$$

$$V = \pm 20\sqrt{3}$$

$$|\tan \alpha| = \frac{20\sqrt{2}}{20}$$

$$\tan \alpha = \sqrt{2}$$

$$\alpha = 55^\circ \text{ (nearest deg.)}$$

$\therefore$ The object strikes the ground at $20\sqrt{3} \text{ ms}^{-1}$ at an angle of 55° .

$$\text{b) Area of segment} = \frac{1}{2}r^2\alpha - \frac{1}{2}r^2\sin\alpha$$

$$A = \frac{1}{2}r^2(\alpha - \sin\alpha)$$

$$\text{If } A=50$$

$$50 = \frac{1}{2}r^2(\alpha - \sin\alpha)$$

$$100 = r^2(\alpha - \sin\alpha)$$

$$\frac{100}{\alpha - \sin\alpha} = r^2$$

$$r = \pm \frac{10}{\sqrt{\alpha - \sin\alpha}}$$

(2)

$$\text{So } r = \frac{10}{\sqrt{\alpha - \sin\alpha}} \text{ as the other is meaningless}$$

$$\text{ie } r = 10(\alpha - \sin\alpha)^{-1/2}$$

$$\text{(ii) } \frac{dr}{dt} = \frac{dr}{d\alpha} \times \frac{d\alpha}{dt}$$

(1)

$$\frac{dr}{d\alpha} = 10 \cdot \frac{1}{2}(\alpha - \sin\alpha)^{-3/2} \cdot (1 - \cos\alpha) = -5 \frac{(1 - \cos\alpha)}{(\alpha - \sin\alpha)^{3/2}}$$

$$\text{When } \alpha = 1.5$$

$$\frac{dr}{d\alpha} = \frac{-5(1.5 - \cos 1.5)}{(1.5 - \sin 1.5)^{3/2}} = -13.044 \text{ (3 dp)}$$

$$\therefore \frac{dr}{dt} = -13.044 \times 0.05$$

$$= -0.65 \text{ (2 dp)}$$

$\therefore$ The radius is decreasing at approx 0.65 cm/s

(1)